

Joe Miguel Robertazzi

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EDUCATION

Stanford University

Stanford, CA

Candidate for Bachelor of Science in Computer Science, Systems Track

Sep. 2023 – June 2027

GPA: 3.9/4.0

Relevant Coursework at Stanford:

- Computer Systems:** Computer Systems Programming, Computer System Architecture, Operating System Design
- Programming:** Data Structures & Algorithms, Parallel Computing, Computer Networking, Embedded Systems
- Artificial Intelligence:** Hardware Accelerators for Machine Learning, Computer Graphics, The Impact of AI

EXPERIENCE

Incoming Software Engineering Intern

June 2026 – Sept. 2026

Apple – Platform Architecture & Performance Control

Cupertino, CA

- Incoming intern joining the Platform Architecture team to develop and optimize system-level software for next-generation hardware performance and power management.

Embedded Systems Intern

June 2025 – Aug. 2025

Stanford University – Computer Systems from the Ground Up Course Infrastructure

Stanford, CA

- Optimized drivers (I2C, SPI, DMA) for 12+ peripherals (OLEDs, cameras, sensors) with improved reliability
- Redesigned I2C/I2S APIs into a unified Hardware Abstraction Layer (HAL), improving system stress-test stability
- Extended platform: ported legacy code to RISC-V, designed custom PCBs, and documented the Vector Extension

Teaching Assistant (CS107E & CS106A/B)

Jan. 2026 – Present

Stanford University – Department of Computer Science

Stanford, CA

- Led bare-metal C labs for 20 students and taught Python and C++ concepts to 10+ students quarterly
- Authored technical guides and helper libraries to enable students to build advanced bare-metal final projects
- Oversaw biweekly grading sessions and debugged complex student programs to provide tailored 1-on-1 feedback

Independent Computational Researcher

Jan. 2021 – June 2023

University of Rochester

Rochester, NY

- Designed 4 modular analysis pipelines in Python (scikit-learn) to process 50M+ avian migration data points
- Generated 100+ geospatial maps (pygmt, Seaborn) correlating magnetic anomalies with population decline
- Published a formal research manuscript results and presented findings at Regeneron STS (Science Talent Search)

PROJECTS

Custom OS Raspberry Pi Trading Tamagotchi | C, Python, Raspberry Pi, UART

Jan. 2026 – Mar. 2026

- Engineered bare-metal Tamagotchi OS linked to Kalshi markets via custom 921600-baud UART-to-Python bridge
- Developed custom drivers: SPI for 128x128 LCD, bit-banged I2C for RTC, and low-latency fast interrupts
- Extended FAT32 filesystem for SD card storage, optimizing UI via instruction caching and ARM overclocking

Bare Metal RISC-V DOOM Port | C, ASM, RISC-V, MangoPi

Jul. 2025 – Aug. 2025

- Ported 1993 DOOM video game to bare metal RISC-V, replacing file I/O with DMA to eliminate OS dependencies
- Engineered drivers for I2C input, I2S-audio, and SPI video with DMA, achieving 20+ fps on the LCD framebuffer
- Implemented a custom C standard library from scratch, extending memory and string routines for bare metal use

Embedded RISC-V Robotic Image Plotter | C, ASM, RISC-V, MangoPi, Python

Jan. 2024 – Mar. 2024

- Delivered a working prototype that prints pixel-accurate images made of LEGOs with a live feedback display
- Engineered a cross-platform data pipeline that serializes images via Python into C structs for bare metal execution
- Implemented image processing in C (downscaling, color quantization) to map pixels to a limited LEGO palette
- Built a full user interface on the MangoPi for image selection, scale, and real-time printing progress tracking

TECHNICAL SKILLS

Programming Languages: C, C++, Assembly (RISC-V/MIPS), Verilog, Python, Java, CUDA, Triton

Developer Tools: Linux Command Line, Conda, AI Copilot, Jupyter Notebook, gdb, Make, Git, AWS

Computing Platforms: MangoPi, RaspberryPi, RTX 4070 & GTX 1080 GPUs, Xilinx PYNQ-Z2 FPGA